CS302

FINAL TERM PREPARATION

BY BRAINY SQUAD 60+ MCQ'S

1. **Counters as the name indicates are not triggered simultaneously**

Select the correct option

- Synchronous
- Negative-Edge triggered
- **Asynchronous**
- Positive-Edge triggered

2. **The terminal count of a 4-bit binary counter in the DOWN mode is_**

Select the correct option

- **0000**
- 0011
- 1111
- 1100

3. **If a circuit suffers "Clock Skew" problem, the output of circuit can't be guarantied.**

Select the correct option

- False
- **True**

4. **The characteristic equation of D-flip-flop implies that**

Select the correct option

- The next state is independent of Inputs
- **The next state is dependent on present state**
- The next state is independent of present state
- The next state is dependent on previous state

5. **Karnaugh map is used in designing**

Select the correct option

- a counter
- a clock
- an UP/DOWN counter
- **All of the above**

6. **A flip-flop is connected to +5 volts and it draws 5 mA of current during its operation, the power dissipation of the flip-flop is**

Select the correct option

- 64 mW
- 10 mW
- 1024 mW
- **25 mW**

7. **The terminal count of a 4-bit binary counter in the UP mode is**

Select the correct option

- 0011
- 1100
- 0000
- **1111**

8. **A divide-by-10 ring counter requires a minimum of**

Select the correct option

- five flip-flops
- four flip-flops
- twelve flip-flops
- **ten flip-flops**

9. **A4-bit UP/DOWN counter is in DOWN mode and in the 1010 state. on the next clock pulse, to what state does the counter go?**

Select the correct option

- **1001**
- 0011
- 1100
- 1011

10. Assume a J-K flip-flop has is on the J and K inputs. The next clock pulse will cause the output to

Select the correct option

- Reset
- **Toggle**
- Latch
- Set

11. According to Moore circuit, the output of synchronous sequential circuit depend/s on of flip flop.

Select the correct option

- **Previous State**
- External Inputs
- Present State
- Next State

12. A 4-bit UP/DOWN counter is in DOWN made and in the 2010 strain on the next clock pulse, to what state does the counter go

Select the correct option

- 1100
- **1001**
- 0011
- 1011

13. The term hold always means

Select the correct option

- Q=0, overline Q = 0
- Q = 0, overline Q = 1
- **No change**
- Q = 1,Q = 0

14. The Synchronous counters are also known as Ripple Counters:

Select the correct option

- **False**
- True

15. The glitches due to "Race Condition" can be avoided by using a

Select the correct option

- Pulse triggered flip-flops
- Negative-Edge triggered flip-flops

- **Positive-Edge triggered flip-flops**
- Gated flip-flops

16. _____ flip-flops are obsolete now.

Select the correct option

- D-Flipflop
- Edge-triggered
- **Master-Slave**
- T-Flipflop

17. Each stage of Master-slave flip-flop works at of the clock signal

Select correct answer

- One fourth
- One third
- **One half**
- Each stage works on complete clock signal

18. The Test Vector definition defines the test vectors for all the three counter inputs and ____ counter output/outputs.

Select the correct option

- Four
- One
- Three
- Two

19. For a down counter that counts from (111 to 000), if current state is "101" the next state will be_____

Select the correct option

- none of given options
- 111
- **110**
- 010

20. Asynchronous mean that...

Select the correct option

- Each flip-flop except the last one is enabled by the output of the preceding flip-flop
- Each alternative flip-flop after the first one is enabled by the output of the preceding flip-flop
- **Each flip-flop after the first one is enabled by the output of the preceding flip-flop**
- Each flip-flop is enabled by the output of the preceding flip-flop

21. _____is said to occur when multiple internal variables change due to change in one input variable

Select the correct option

- **Race condition**
- Clock Skew
- Hold delay

- Hold and Wait

22. To parallel load a byte of data into a shift register, there must be

Select the correct option

- eight clock pulse
- one clock pulse for each 0 in the data
- **one clock pulse**
- one clock pulse for each 1 in the data

23. The counter states of the range of numbers of a counter is determined by the formula. ("n" represents the total number of flip-flops)

Select the correct option

- (2 raise to power n)
- (n raise to power 2 and then minus 1)
- **(2 raise to power n and then minus 1)**
- (n raise to power 2)

24. The 74HC163 is a 4-bit Synchronous Counter. It has.....data output pins

Select the correct option

- 2
- 6
- **4**
- 8

25. Each stage of Master-slave flip-flop works at of the clock signal

Select the correct option

- Each stage works on complete clock signal
- **One half**
- One fourth
- One third

26. Choose the best possible answer of following question:

The D flip-flop is only activated by_____.

Select the correct option

- **a positive edge trigger**
- a negative edge trigger
- both positive & negative edge trigger
- None of the given options

27. The ____ inputs can be directly mapped to Karnaugh maps.

Select the correct option

- **S-R**
- J-K
- External
- Flip-flop

28. In designing any counter the transition from a current state to the next sate is determined by

Select the correct option

- Only inputs
- Only current state
- current state and outputs
- **Current state and inputs.**

29. Karnaugh map is used in designing

Select the correct option

- **All of the above**
- a clock
- an UP/DOWN counter
- a counter

30. With a 100 KHz clock frequency, eight bits can be serially entered into a shift register in

Select the correct option

- 10 micro seconds
- **80 micro seconds**
- 8 micro seconds
- 80 mili seconds

31. Which of the following is an invalid BCD code?

Select the correct option

- 1001
- 0011
- **1101**
- 0101

32. Reflected Binary Code is also known as ..... code.

Select the correct option

- BCD
- ASCII
- **Gray Code**
- Excess-3

33. Convert the gray code 1011 to binary

Select the correct option

- **1101**
- 0111
- 1011
- 1010

34. Three decade counter would have

Select the correct option

- 2 BCD counters
- **3 BCD counters**
- 5 BCD counters
- 4 BCD counters

35. Which one is a basic comparator?

Select the correct option

- OR
- NAND
- **XOR**
- AND

36. The parallel outputs of a counter circuit represent the

Select the correct option

- Parallel data word
- Counter modulus
- **Clock count**
- Clock frequency

37. TTL 74LS85 is a

Select the correct option

- 8-bit magnitude comparator
- 8-bit word comparator
- **4-bit magnitude comparator**
- 1-bit digital comparator

38. To create a Johnson Counter by using D-flip flops, how flip flops should be connected together?

Select the correct option

- Q' output of previous flip flop should be connected to clock input of second flip flop
- Q output of previous flip flop should be connected to D input of second flip flop
- Q output of previous flip flop should be connected to clock input of second flip flop
- **Q' output of previous flip flop should be connected to D input of second flip flop**

39. Ripple counters are also called

Select the correct option

- VLSI counters
- **Asynchronous counters**
- Synchronous counters
- SSI counters.

40. A decimal counter has ____ states.

Select the correct option

- 2
- **10**
- 9
- 7

41. Three decade counter would have

Select the correct option

- 2 BCD counters
- **3 BCD counters**
- 5 BCD counters
- 4 BCD counters

42. An identify comparator is defined as a digital comparator which has

Select the correct option

- No output terminals
- Two output terminals
- Three output terminals
- **Only one output terminal**

43. What is the maximum possible range of bit-count specifically in n-bit binary counter consisting of 'n' number of flip-flops?

Select the correct option

- **0 to 2^n-1**
- 0 to 2^n+1
- 0 to 2^n
- 0 to 2^n+1

44. **Comparators are used in… .. … ..**

Select the correct answer

- **CPU**
- Memory
- Motherboard
- Hard drive

45. Which of the following is the modified code of Excess-3 code?

Select correct option

- ASCII
- BCD
- EBDIC
- **Gray Code**

46. State of flip-flop can be switched by changing its

Select the correct option

- **Input Signal**
- Contemporary Signal
- Momentary Signal
- Output Signal

47. The CONSTATE CLK Clock is used to indicate that the state variables change on a clock transition

- Select the correct option

- FLOOR
- OPEN
- **CONSTATE**
- MOTION

48. With a 100 KHz clock frequency, eight bits can be serially entered into a shift register in

Select the correct option

- **80 micro seconds**
- 80 mili seconds
- 8 micro seconds
- 10 micro seconds

49. The 64-cell array organized as 8x8 cell array is considered

Select the correct option

- As an 4 byte memory
- As an 18 byte memory
- **As an 8 byte memory**
- As an 64 byte memory

50. Two types of memories namely the first in-first out (FIFO) memory and last in first out (UFO) are implemented using

Select the correct option

- **Shift Register**
- Reduce Registers
- Ring Buffers
- Circular Buffers

51. The normal data inputs to a flip-flop (D. 5 and R. and K. T) are referred to as inputs

Select the correct option

- Combinational
- **Synchronous**
- Asynchronous
- Sequential

52. In mealy machine the output depends on

Select the correct option

- The input
- None of the above
- **The current state**
- Current state and the inputs

53. The Static Ram (SRAM) is non-volatile and is not a density memory as a latch is required to store a single bit of information

Select the correct option

- Medium
- Hot
- **High**
- Low

54. Two states are said to be equal if they have exactly some

Select the correct option

- Next state
- Outputs
- None of the given options
- **Inputs**

55. The ROM used by a computer is relatively very _____ as it stores few bytes of code used to Boot the Computer system on power

Select the correct option

- High
- heavy
- **Small**
- Large

56. Smallest unit of data is

Select the correct option

- A 16-bits word
- A4-bits nibble
- **A bit**
- An 8-bits word

57. Which of the following is a volatile memory?

- EPROM
- EEPROM
- PRO
- **DRAM**

58. Which of the following is the drawback of DRAM?

Select the correct option

- **Discharging of the capacitor aver a period of time.**
- Extra circuitry is required to refresh the capacitor
- All of the above are true
- All the information stored in terms of binary bits would be lost capacitor is not recharged

59. Implementation of the FIFO buffer in _____ is usually takes the form of a circular buffer

Select the correct option

- **RAM**
- PPROM
- Flash Memory
- ROM

60. Number of states in an 8-bit Johnson counter sequence are

Select the correct option

- 8
- 12
- 14
- **16**

61. when the transmission line is idle in an asyrichronous transmission

Select the correct option

- It remains in previous state
- It is set to logic low
- **It is set to logic high**
- State of transmission line is not used to start transmission

62. Two State variables allow a maximum of states

Select the correct option

- **Four**
- Two
- Sixteen
- Eight

63. A memory organized to store nibble date values requires a _____ Wide data bus

Select the correct option

- 2-bit
- **4-bit**
- 8-bit
- 16-b

64. Memory is arranged in

Select the correct option

- Linear fashion
- Three-dimensional manner
- **Two-dimensional manner**
- Random fashion

65. The duration for which the elevator doors are opened, and remain open, and time it takes for the elevator to move from one floor to/an……

Select the correct option

- None of the given options
- **Clock Signal**
- Input Signal
- Output Signal

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